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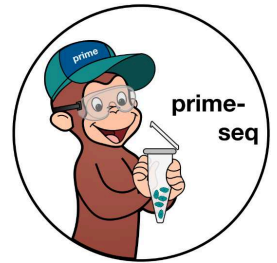
Version 2

prime-seq V.2

 [Scientific Reports](#)

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Aleksandar Janjic¹, Lucas Esteban Wange¹, Johannes JWB Bagnoli¹, Johanna Geuder¹, Phong Nguyen¹, Daniel Richter¹, Christoph Ziegenhain², Beate Vieth¹, Wolfgang Enard¹

¹Ludwig-Maximilians-Universität München; ²Karolinska Institute Stockholm

Ag Enard

Epi Aging Lab

1 more workspace



Aleksandar Janjic

Ludwig-Maximilians-Universität München

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We use this protocol and it's working

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Abstract

Cost-efficient library generation by early barcoding has been central in propelling single-cell RNA sequencing. Here, we optimize and validate prime-seq, an early barcoding bulk RNA-seq method. We show that it performs equivalently to TruSeq, a standard bulk RNA-seq method, but is fourfold more cost-efficient due to almost 50-fold cheaper library costs. We also validate a direct RNA isolation step, show that intronic reads are derived from RNA, and compare cost-efficiencies of available protocols. We conclude that prime-seq is currently one of the best options to set up an early barcoding bulk RNA-seq protocol from which many labs would profit.

Guidelines

- All reagents and plastic-ware can be found in the 'Materials' section.
- Use only RNase free supplies and clean all surfaces and tools with RNase Away prior to working
- Make sure all steps involving cell lysate and RNA before reverse transcription are carried out swiftly and on ice.
- All primer sequences are listed below:

A	B	C	D	E	F
Oligo	Vendor	Purification	Working Conc.	Sequence	Notes
Barcoded Oligo-dT (E3V7NEXT)	Sigma	Cartridge	10 µM	ACACTCTTTCCCTACACGAC GCTCTTCCGATCT[12 bp BC]NNNNNNNNNNNNNNNN VTTTTTTTTTTTTTTTTTTT TTTTTTTTTTVN	
Template Switching Oligo (TSO) (E5V7NEXT)	Sigma	RNase-Free HPLC	100 µM	Biotin- ACACTCTTTCCCTACACGAC GCrGrGrG	
Preamp Primer (SINGV6)	Sigma	Standard Desalting	10 µM	Biotin- ACACTCTTTCCCTACACGAC GC	
3' enrichment primer (P5NEXTPT5)	Sigma	Standard Desalting	5 µM	AATGATACGGCGACCACCG AGATCTACACTCTTTCCCTA CACGACGCTCTTCCGATCT	
i7 Index Primer (Nextera)	IDT	Trugrade	5 µM	CAAGCAGAAGACGGCATA GAGAT[i7]GTCTCGTGGGCT CGG	
i5 Index Primer (TruSeq)	IDT	Trugrade	5µM	AATGATACGGCGACCACCG AGATCTACAC[i5]ACACTCTT TCCCTACACGACGCTCTTC CGATCT	
prime-seq Adapter AntiSense	IDT	Standard Desalting	1.5 µM	/5Phos/CTGTCTCTTATACAC ATCT	Duplexed DNA
prime-seq Adapter Sense	IDT	Standard Desalting	1.5 µM	GTCTCGTGGGCTCGGAGAT GTGTATAAGAGACAGT	Duplexed DNA

Specific barcoded oligodT (E3V7NEXT) sequences:



E3V7_Set1.txt



E3V7_Set2.txt



Materials

MATERIALS

- ✕ DNase I Reaction Buffer - 6.0 ml **New England Biolabs Catalog #B0303S**
- ✕ DNase I (RNase-free) - 1,000 units **New England Biolabs Catalog #M0303S**
- ✕ Deoxynucleotide Solution Mix - 40 umol of each **New England Biolabs Catalog #N0447L**
- ✕ Exonuclease I (E.coli) - 3,000 units **New England Biolabs Catalog #M0293S**
- ✕ Quant-it™ PicoGreen® dsDNA Assay Kit **Life Technologies Catalog #P7589**
- ✕ β -mercaptoethanol **Merck MilliporeSigma (Sigma-Aldrich) Catalog #M3148**
- ✕ QuantiFluor(R) RNA System **Promega Catalog #E3310**
- ✕ Proteinase K solution, 20 mg ml – 1 **Ambion Catalog #AM2546**
- ✕ 5 M Sodium chloride (NaCl) **Merck MilliporeSigma (Sigma-Aldrich) Catalog #S5150-1L**
- ✕ Agilent High Sensitivity DNA Kit **Agilent Technologies Catalog #5067-4626**
- ✕ Buffer RLT Plus **Qiagen Catalog #1053393**
- ✕ Maxima H Minus Reverse Transcriptase (200 U/uL) **Thermo Fisher Scientific Catalog #EP0752**
- ✕ NEBNext Ultra II FS DNA Library Prep with Sample Purification Beads - 24 rxns **New England Biolabs Catalog #E6177S**
- ✕ EDTA **Merck MilliporeSigma (Sigma-Aldrich) Catalog #E7889**
- ✕ Ethanol absolute **Carl Roth Catalog #9065.4**
- ✕ Igepal **Merck MilliporeSigma (Sigma-Aldrich) Catalog #I8896**
- ✕ KAPA HiFi 2x RM **Kapa Biosystems Catalog #KR0370**
- ✕ Poly(ethylene glycol) **Merck MilliporeSigma (Sigma-Aldrich) Catalog #89510**
- ✕ UltraPure DNase/RNase Free Distilled Water **Catalog #10977-049**
- ✕ Trizma hydrochloride solution **Merck MilliporeSigma (Sigma-Aldrich) Catalog #T2694**
- ✕ Aluminium seals for cold storage **Catalog #391-1275**
- ✕ Filter tips 96 low retention 10 uL **Catalog #771265**
- ✕ PCR Seals **Thermo Scientific Catalog #AB0558**
- ✕ twin.tec 96-well DNA LoBind Plates **Eppendorf Catalog #0030129504**
- ✕ Sera-Mag Speed Beads **GE Healthcare Catalog #65152105050250**
- ✕ Sodium Azide **Merck MilliporeSigma (Sigma-Aldrich) Catalog #S2002-100G**

Safety warnings

 Please follow all Manufacturer safety warnings and recommendations.

Before start

Wipe bench surfaces with RNase Away and keep working environment clean.



Preparation

12m

- 1 Clean all surfaces and pipettes with RNase Away 5m
- 2 Thaw frozen buffers and primers on ice 10m
- 3 Prepare 80% EtOH (approximately 45 mL for 96 samples) 2m
- 4 **When running the protocol for the first time prepare Cleanup Beads (see end of the protocol)!** 45m
- 5 prime-seq can be used on lysate or extracted RNA. It is essential, however, that the samples either have the same input or that they are normalized after the RNA is extracted, otherwise sequencing depth per sample will be impacted. Based on your starting material, please follow one of the following cases: 

STEP CASE

Lysate (similar input), Direct Lysis 106 steps

Follow this case if you are testing samples that have **similar input** (i.e. the expected RNA amount is the same between samples). The steps here will guide you in digesting residual proteins in your samples, extracting the RNA, digesting DNA, preparing RNA-seq libraries, and finally sequencing.

Example: investigating the genotype effect on transcription in 5,000 neurons

First Time Setup

- 6 When running the direct lysis protocol for the first time, prepare Bead Binding Buffer (see end of the protocol)!

Sample Collection

- 7 Prepare **Lysis Buffer** according to the number of samples. 2m

Reagent	Well	Plate
---------	------	-------



	RLT Plus Buffer	99 μ L	10.89 mL
	β -mercaptoethanol	1 μ L	110 μ L
	Total	100 μL	11 mL

Note

If sample volume exceeds 25 % of total lysate, use 2x TCL buffer (Qiagen, #1070498) + 1 % β -mercaptoethanol

8 Add  100 μ L of **Lysis Buffer** to each well of a semi-skirted 96-well PCR plate

1m

9 Add cells or tissue to wells



Note

Cells

Minimum: 100 cells, Optimum: 10,000 cells

Make sure that the same number of cells are used for each sample. Large differences between cells will impact distribution of sequencing reads and can potentially affect normalization.

**Note****Tissue**

If samples are difficult to lyse they should be homogenized using a tissue homogenizer.

Tissue should be a relatively small and not exceed more than 1000 ng of RNA. Tissue samples should be normalized by weight and be the same type of tissue.

Large differences between tissue samples will impact distribution of sequencing reads and can potentially affect normalization.

If you are unsure if the samples will contain the same amount of RNA, it is best to switch to the "Lysate (*variable*)" case in Step 13.

- 10 Transfer  50 μL of **lysate** to a new plate, return one plate immediately to -80 C freezer to save as a backup

1m

Note

Conversely, one can prepare two plates during sorting with 50 μL of lysis buffer.

Proteinase K Digest

30m

- 11 Add  1 μL **Proteinase K** (20 mg/mL) and  1 μL **EDTA** (25 mM) to each well

1m

- 12 Incubate for  00:15:00 at  50 °C and then heat inactivate the Proteinase K for  00:10:00 at  75 °C

25m

Bead Clean Up

20m



13 Mix each bulk sample (50 μ L per well) with  100 μ L of **Cleanup Beads (22% PEG)**

1m

14 Incubate for  00:05:00 at  20 °C (Room Temp)

5m

Note

While binding, prepare **DNase I Mix** (Step 28)

15 Place on magnet stand until clear (~3 min) and then discard supernatant

3m

16 Wash with  100 μ L of **80% EtOH** while the plate is on the magnet. Discard the supernatant

2m

Note

After adding EtOH, incubate for 30 s so that all beads are bound to magnet.

17 Repeat wash step once more

2m

18 Air dry beads for  00:03:00

3m

Note

Depending on temperature and humidity, the beads may dry faster. Therefore it is important to regularly check the beads and avoid over-drying.

DNase I Digest

1m



19 Add  5 μL H₂O and **resuspend beads by vortexing vigorously**

2m

Note

If you encounter bead clumping at this step, try to resuspend the beads by vigorously pipetting the samples. We have generated high quality prime-seq libraries despite heavy clumping.

20 Prepare **DNase I Mix**

3m

Reagent	Well	Plate
DNase I	1 μL	110 μL
DNase I Buffer (10x)	2 μL	220 μL
Bead Binding Buffer (2x)	10 μL	1.1 mL
H ₂ O	2 μL	220 μL
Total	15 μL	1.65 mL

21 Add  15 μL of **DNase I Mix** and mix by pipetting

2m

22 Incubate DNase I Mix and beads for  00:10:00 at  20 °C (Room Temp)

10m

23 Heat inactivate the DNase I by adding  1 μL of **EDTA (100 mM)** and incubating for  00:05:00 at  65 °C

6m



24 Place plate on magnet stand until clear (~3 min) and discard the supernatant.

3m

25 Wash with  100 μL of **80% EtOH** while the plate is on the magnet. Discard the supernatant

2m

26 Repeat wash step once more

2m

27 Air dry beads for  00:05:00

5m

Note

Depending on temperature and humidity, the beads may dry faster. Therefore it is important to regularly check the beads and avoid over-drying.

Note

While drying, prepare **Reverse Transcription Mix**.

Reverse Transcription

5m

28 Prepare **Reverse Transcription Mix**

5m

	Reagent	Well	Plate
	Maxima H Minus RT	0.15 μL	16.5 μL
	Maxima RT Buffer (5x)	2 μL	220 μL
	dNTPs (25 mM)	0.4 μL	44 μL
	TSO (E5V7NEXT) (100 μM)	0.1 μL	11 μL
	UltraPure Water	2.35 μL	258.5 μL
	Total	5 μL	550 μL



29 Add  4 μL **H₂O** to the beads

1m

Note

The 4 μL of water can be combined with the Reverse Transcription Mix by increasing the water in Row 6 from 2.35 μL to 6.35 μL .

If working with many samples, or if using a stepper pipette or robot, we find that it is better to add some water separately to prevent the beads from drying too much.

30 Add  5 μL **Reverse Transcription Mix**

1m

31 Add  1 μL of **Barcoded oligodT (E3V7NEXT) (10 μM)** per well

2m

32 Incubate for  01:30:00 at  42 °C

1h 30m

cDNA Pooling & Purification

5m

33 Place the plate on a magnet

3m

34 Pool the supernatant of all wells into a 2 mL tube

10m

35 Add  10 μL of **Cleanup Beads (22% PEG)** *per sample* for a 1:1 ratio (e.g. 240 μL for 24 samples)

5m

Note

The EDTA in the **Cleanup Beads (22% PEG)** will inactivate the RT.

36 Incubate for  00:05:00 at  Room temperature to allow binding of the cDNA onto beads

5m



37 Place the tube on the magnet stand until clear (~3 min) and discard supernatant

3m

38 Wash with  1 mL of **80% EtOH** while the tube is on the magnet, discard the supernatant

1m

Note

Volume of EtOH should be adjusted depending on the number of samples. More samples will require more EtOH to cover the beads completely.

39 Repeat wash step once more

1m

40 Air dry beads for  00:05:00

5m

Note

Depending on temperature and humidity, the beads may dry faster. Therefore it is important to regularly check the beads and avoid over-drying.

41 Elute the beads in  17 μ L of **UltraPure Water**

1m

42 Incubate for  00:05:00 at RT and transfer to a new PCR tube or plate

5m

Exonuclease I Treatment

35m

43 Add  2 μ L of **Exol Buffer (10x)** and  1 μ L of **Exonuclease I**. Incubate as follows:

35m

	Step	Temperature	Time
	Incubation	37 C	20 min



Heat Inactivation	80 C	10 min
Storage	4 C	∞

- 44 Mix each sample (20 μ L per well) with  16 μ L of **Cleanup Beads (22% PEG)** for a 1:0.8 ratio 1m
- 45 Incubate for  00:05:00 at  Room temperature to allow binding of the cDNA onto beads 5m
- 46 Place the tube on the magnet stand until clear (~3 min) and discard supernatant 3m
- 47 Wash with  50 μ L of **80% EtOH** while the tube is on the magnet, discard the supernatant 1m
- 48 Repeat wash step once more 1m
- 49 Air dry beads for  00:05:00 5m

Note

Depending on temperature and humidity, the beads may dry faster. Therefore it is important to regularly check the beads and avoid over-drying.

- 50 Elute the beads in  20 μ L of **UltraPure Water** 1m
- 51 Incubate for  00:05:00 at RT and transfer to a new PCR tube or plate 5m

Full length cDNA Amplification

- 52 Prepare **Pre Amplification Mix** 1m



Reagent	1x
KAPA HiFi 2x RM	25 μ L
Pre-amp Primer (SINGV6) (10 μ M)	3 μ L
UltraPure Water	2 μ L
Total	30 μL

53 Add  30 μ L **Pre Amplification Mix** to sample

1m

54 Incubate the Pre Amplification PCR as follows:

1h 30m

Step	Temperature	Time	Cycles
Initial Denaturation	98 C	3 min	1 cycle
Denaturation	98 C	15 sec	10 cycles*
Annealing	65 C	30 sec	
Elongation	72 C	4 min	
Final Elongation	72 C	10 min	1 cycle
Storage	4 C	∞	

**Note**

Adjust the number of cycles based on input (sample number, cell number, or concentration).

As a rule of thumb we assume big cells like embryonic stem cells to contain 10 pg of total RNA and small cells like T-cells ~ 1-2 pg

As a general guide we recommend:

	Total RNA Input	Cycles
	10 ng	16
	50 ng	14
	100 ng	12
	500 ng	10
	1000 ng	9

cDNA Bead Purification

1m

55 Mix sample with  40 µL **Clean Up Beads (22% PEG)** for a ratio of 1:0.8

1m

56 Incubate for  00:05:00 at  20 °C (Room Temp)

5m

57 Place the tube on the magnet stand until clear (~3 min) and discard supernatant

3m

58 Wash with  100 µL of **80% EtOH** while the tube is on the magnet, discard the supernatant

1m

59 Repeat wash step once more

1m

60 Air dry beads for  00:05:00

5m

**Note**

Depending on temperature and humidity, the beads may dry faster. Therefore it is important to regularly check the beads and avoid over-drying.

61 Elute cDNA in  10 μ L **UltraPure Water**

1m

62 Incubate for  00:05:00 at RT and transfer to a new PCR tube or plate

5m

Note

Stopping Point. Samples can be safely stored at  -20 °C and protocol can be continued at a later date.

**cDNA Quantification and Quality Check**

45m

63 Quantify the cDNA using the **Quant-iT PicoGreen dsDNA assay kit** or equivalent Qubit following the manufacturer's protocol. Use 1 μ L of clean cDNA for quantification.

10m



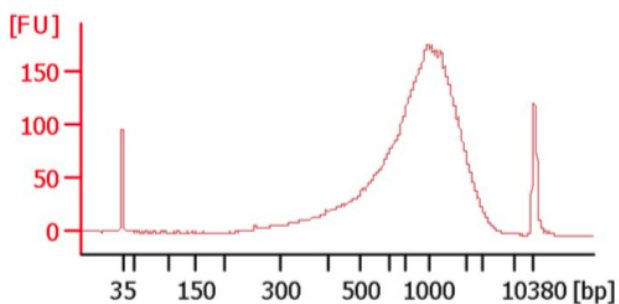
64 Quality check the cDNA using the Agilent 2100 Bioanalyzer with **High Sensitivity DNA Analysis Kits**.

45m

Note

Passing the cDNA quality check does not guarantee that the data will be of high quality, however, if the cDNA fails the quality check it will usually not yield good libraries and will therefore generate lower quality data.

Expected result



Library Preparation

65



Note

Before starting, read the library preparation section carefully as there are a few steps that are very time sensitive.

66 Prepare **Fragmentation Mix**

1m

Reagent	1x
Ultra II FS Reaction Buffer	1.4 μ L
Ultra II FS Enzyme Mix	0.4 μ L
cDNA (4-8 ng/ μ L)	2.5 μ L
TE	1.7 μ L
Total	6 μL

**Note**

Ensure that the Ultra II FS Reaction Buffer is completely thawed. If a precipitate is seen in the buffer, pipette up and down several times to break it up, and quickly vortex to mix. Place on ice until use.

Note

Vortex the Ultra II FS Enzyme Mix for 5-8 seconds prior to use for optimal performance.

67 Vortex the **Fragmentation Mix** for  00:00:05 and immediately proceed to step 67

10s

68 Incubate the Fragmentation reaction as follows:

40m

	Step	Temperature	Time
	Pre-Cool	4 C	∞
	Fragmentation	37 C	5 min
	A Tailing and Phosphorylation	65 C	30 min
	Storage	4 C	∞

Note

Set heated lid to 75° C. Make sure the lid is at the correct temperature before you start the reaction.
Skip the first incubation step once you have added your samples.

Adapter Ligation

20m

69 Prepare **Adapter Ligation Mix**

1m



Reagent	1x
NEBNext Ultra II Ligation Master Mix	6 µL
NEBNext Ligation Enhancer	0.2 µL
prime-seq Adapter (1.5 µM)	0.5 µL
Total	6.7 µL

70 Add  6.7 µL **Adapter Ligation Mix** to each replicate

1m

71 Incubate for  00:15:00 at  20 °C

15m

Note

Turn off heated lid

72 Add  37.3 µL Buffer EB to Samples

1m

73 Mix Sample with  26 µL **SPRI select beads**

1m

Note

We use SPRI Select Beads here instead of our home made 22% Clean Up beads for their guaranteed QCed size selection properties.

Note

The volume of SPRI select beads used during library size selection can be adjusted based on desired library size. Optimization for your samples may be required.



74 Incubate for 00:05:00 at 20 °C (Room Temp) 5m

75 Place the plate on the magnet stand until clear and **transfer** 76 μL **supernatant to clean well.** 3m

Note

Be careful not to discard! This is your sample!

76 Mix supernatant with 10 μL **SPRI select beads** 1m

Note

The volume of SPRI select beads used during library size selection can be adjusted based on desired library size. Optimization for your samples may be required.

77 Incubate for 00:05:00 at 20 °C (Room Temp) 5m

78 Place the plate on the magnet stand until clear and discard supernatant 3m

79 Wash with 150 μL of **80% EtOH** while the plate is on the magnet, discard the supernatant 1m

80 Repeat wash step once more 1m

81 Air dry beads for 00:05:00 5m

Note

Depending on temperature and humidity, the beads may dry faster. Therefore it is important to regularly check the beads and avoid over-drying.

82 Elute samples in  11 μL **0.1X TE** (dilute 1X TE Buffer 1:10 in water) for  00:05:00

5m

Library PCR

45m

83 Transfer  10.5 μL of samples to clean wells

84 Add  1 μL of **Index Primer (Nextera i7, 5 μM)** to each well

Note

This is the unique index that will be used for demultiplexing libraries.

85 Add  1 μL of **Index Primer (TruSeq i5, 5 μM)** to each well

Note

Alternatively the universal primer P5NEXTPT5 can be used in case the second index will not be sequenced.

86 Prepare **Library PCR Mix** by adding  12.5 μL

Note

Although scaled down, there will not be sufficient Q5 Master Mix (M0544L) in the kit. This item will have to be ordered separately.

87 Incubate the **Library PCR** reaction as follows:

Step	Temperature	Time	Cycles
Initial Denaturation	98 C	30 sec	1 cycle



	Denaturation	98 C	10 sec	10 cycles*
	Annealing/Elongation	65 C	1 min 15 sec	
	Final Elongation	65 C	5 min	1 cycle
	Storage	4 C	∞	

Note

Adjust the number of cycles based on cDNA input.

As a general guide we recommend:

	cDNA Input	Cycles
	20 ng	10
	10 ng	11
	5 ng	12

Double Size Selection

25m

88 Add  25 µL Buffer EB to Index PCR

89 Mix Index PCR with  26 µL **SPRI select beads**

Note

We use SPRI Select Beads here instead of our home made 22% Clean Up beads for their guaranteed QCed size selection properties.

**Note**

The volume of SPRI select beads used during library size selection can be adjusted based on desired library size. Optimization for your samples may be required.

90 Incubate for 00:05:00 at 20 °C (Room Temp)

91 Place the plate on the magnet stand until clear and **transfer** 76 µL **supernatant to clean well.**

Note

Be careful not to discard! This is your library.

92 Mix supernatant with 10 µL **SPRI select beads**

Note

The volume of SPRI select beads used during library size selection can be adjusted based on desired library size. Optimization for your samples may be required.

93 Incubate for 00:05:00 at 20 °C (Room Temp)

94 Place the plate on the magnet stand until clear and discard supernatant.

95 Wash with 150 µL of **80% EtOH** while the plate is on the magnet, discard the supernatant

96 Repeat wash step once more

97 Air dry beads for 00:05:00

**Note**

Depending on temperature and humidity, the beads may dry faster. Therefore it is important to regularly check the beads and avoid over-drying.

98 Elute in  15 μ L **UltraPure Water**.

99 Incubate for  00:05:00 and then place on magnet until clear. Transfer eluted library to new well.

Note

Stopping point. The libraries can be safely stored at  -20 °C until they will be QCed and sequenced.

QC and quantification

45m

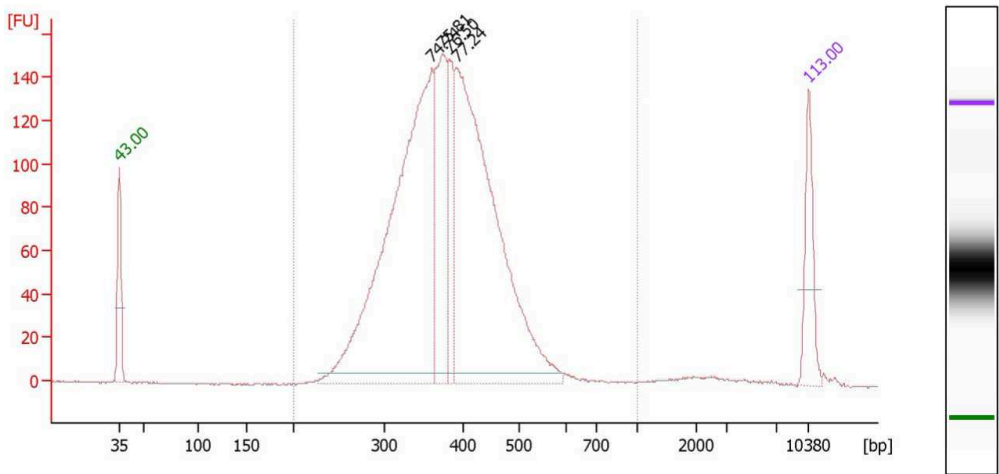
100 Quantify and quality control the library using the Agilent 2100 Bioanalyzer with **High Sensitivity DNA Analysis Kits**.

Note

Bulk libraries often yield high concentrations, which should be diluted to get accurate molarity measurements on the Bioanalyzer. Ideally, do not load more than 2 ng onto the chip.

Expected result

Libraries will typically exceed 1-5 ng/ μ l concentration



Sequencing

1m

101 Samples should be submitted according to your Sequencing Facility specifications. prime-seq is compatible with Illumina Sequencing.

At least 8 cycles are required for the Index Read (i7) and 28 cycles for the Read 1 (BC+UMI). Dual index sequencing can be done when using patterned flowcells. Read 2 (DNA) should be adjusted based on the quality of the genome being mapped to, but for human and mouse 50 cycles is sufficient.

Some potential sequencing options:

	A	B	C	D	E	F
	Sequen cer	Read 1	Read 2	Index Read (i7)	Index Read (i5)	Kit
	NovaSe q	28	94	8	8	NovaSeq SP v1.5 100 cycle



	A	B	C	D	E	F
	NextSeq q 500/550	28	56	8	0	NextSeq 500/550 HiOut v3 75 cycle
	NextSeq q 1000/2000	28	94	8	8	NextSeq 1000/2000 P2 100 cycle
	NextSeq q 2000	28	52	8	0	NextSeq 2000 P3 50 cycles
	HiSeq	28	114	8	0	HiSeq 3000/4000 150 cycles

Prepare Cleanup Beads (22% PEG)

10m

102 Prepare **PEG Solution (22%)** by adding all ingredients to a 50 mL falcon tube

10m

	Reagent	Amount
	PEG 8000	11 g
	NaCl (5M)	10 mL
	Tris-HCl (1M, pH 8.0)	500 µL
	EDTA (0.5M)	100 µL
	IGEPAL (10% solution)	50 µL
	Sodium Azide (10% solution)	250 µL
	UltraPure Water	up to 49 mL
	Total	49 mL

Note

Do not add the total amount of water until after PEG is completely solubilized



- 103 Incubate at 40°C and vortex regularly until PEG is completely dissolved 10m
- 104 Resuspend **Sera-Mag Speed Beads** carefully and pipette $1000\ \mu\text{L}$ of bead suspension into a 1.5 mL tube 1m
- 105 Place on magnet stand and remove storage buffer 1m
- 106 Add $1000\ \mu\text{L}$ of **TE Buffer** (10 mM Tris-HCl, pH 8.0, 1 mM EDTA) and resuspend beads 30s
- 107 Place on magnet stand and remove supernatant 30s
- 108 Repeat wash step one more time 1m
- 109 Add $900\ \mu\text{L}$ **TE Buffer** (10 mM Tris-HCl, pH 8.0, 1 mM EDTA) and resuspend beads 30s
- 110 Add the washed **Sera-Mag Speed Beads** to the **PEG Solution (22%)** and mix well 1m 30s

Note

The final **Cleanup Beads (22% PEG)** can be aliquoted and stored at 4°C for up to six months

Prepare Bead Binding Buffer

10m

- 111 Prepare **Bead Binding Buffer (2x)** 10m

Reagent	
PEG 8000	1.1 g



	NaCl (5 M)	1 mL
	Tris-HCl (1 M, pH 8.0)	50 µL
	Igepal (10% solution)	5 µL
	Sodium Azide (10% solution)	25 µL
	H ₂ O	to 5 mL
	Total	5 ml

Note

The **Bead Binding Buffer (2x)** can be stored at  Room temperature for up to six months.